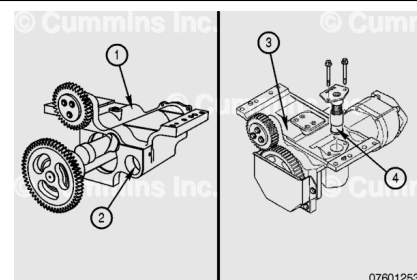


Trainings ▾

General Information

There have been two styles of lubricating oil pump high pressure relief valve used on QSK45 and QSK60 engines. On earlier lubricating oil pumps (1) the high pressure relief valve was positioned horizontally, relief valve position shown (2). On current pumps (3), the high pressure relief valve is positioned vertically, relief valve position shown (4). Lubricating oil pumps with the high pressure valve in the horizontal position were made obsolete in early 2003 as per the Engine Serial Number (ESN) numbers below.



ESN first 33153112 for QSK45 and ESN first 33154182 for QSK60.

For the purposes of this procedure, the two high pressure relief valves will be referred to as horizontal or vertical position.

Note : Some engines built after the ESN first highlighted above were built with lubricating oil pumps that did not contain a high pressure relief valve. All engines after ESN first 33164296 for QSK45 and ESN first 33164290 for QSK60 were built with lubricating oil pumps that contain a high pressure relief valve in the vertical position (2).

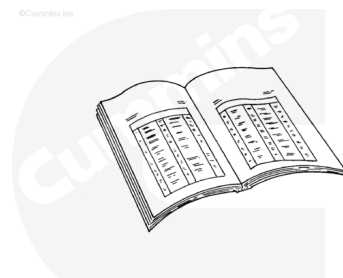
Note : The lubricating oil pump shown in this procedure can possibly be different in appearance from the pump actually installed. Although different in appearance, the procedure remains the same.

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the lubricating oil pump. Refer to Procedure 007-031 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)

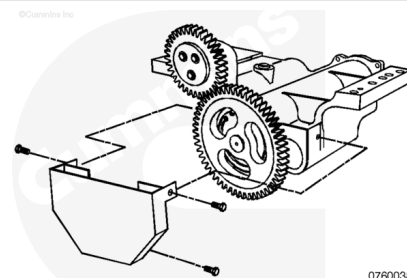


ck800wa

Remove

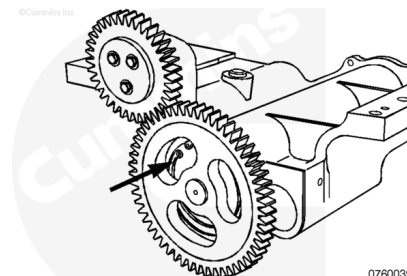
Horizontal Position

Remove the three oil pump gear cover mounting capscrews and the oil pump gear cover.



07600389

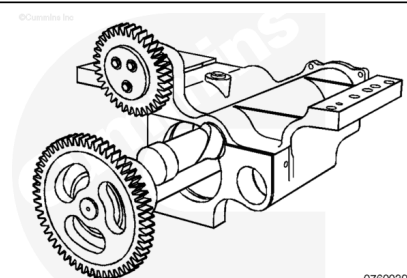
Remove the main rotor bearing retaining snap ring.



07600390

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



07600391

⚠ CAUTION ⚠

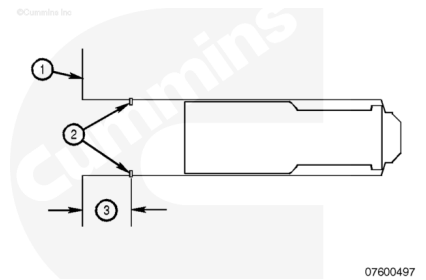
The oil pump must be properly supported to prevent falling and damaging the oil pump.

Slide the rotor assembly forward 152 to 203 mm [6 to 8 in].

Do **not** remove the rotor assembly unless damage is seen.

Before removing the high pressure relief valve, perform the following measurement to determine the spool beat-in wear.

Use a depth gauge to measure the distance (3) from the machined face (1) of the lubricating oil pump to the front of the snap ring (2).

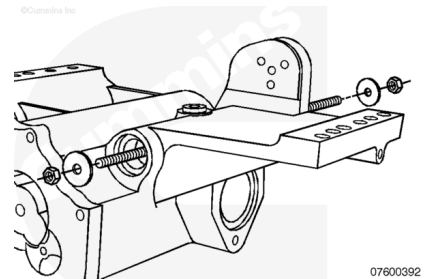


Install a 5/16-inch x 10-inch threaded rod through the center of the high pressure relief valve assembly.

The threaded rod **must** protrude all of the way through the high pressure relief valve assembly.

Install a 32 mm [1.25 in] washer and nut on the end of the threaded rod at the rear of the oil pump.

Install a washer that has a smaller diameter than the snap ring and a nut on the other end of the threaded rod.

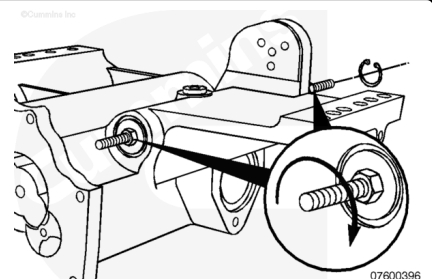


Tighten the nuts until the spring is compressed enough to allow the snap-ring to be removed.

Remove the snap-ring.

Remove the threaded rod, spring retainer, spring, and the high pressure relief valve from the lubricating oil pump.

Remove the spring retainer, spring, and high pressure relief valve from the threaded rod.

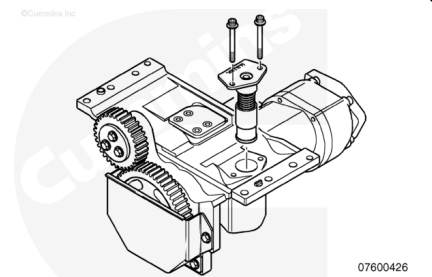


Vertical Position

Note : Care **must** be taken when removing the pressure relief valve capscrews. The compressed spring is under high compression and can cause damage if not removed carefully.

Note : The capscrews have an extended length to aid removal.

Remove the capscrews and the pressure relief valve assembly.



Clean and Inspect for Reuse

Horizontal Position

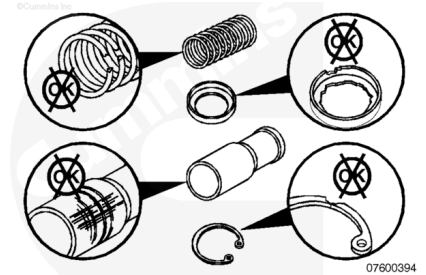
Inspect the spring, spring retainer, and snap-ring for damage.

If any damage is found, the damaged part **must** be replaced.

Inspect the spool valve for scuffing or scoring.

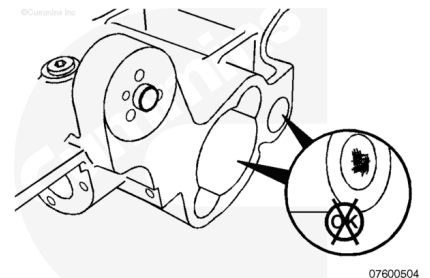
If any damage is found, inspect the high pressure relief valve bore in the oil pump.

Note : Polished areas on the spool valve is normal wear.



Inspect the lubricating oil high pressure regulator bore for scuffing or other damage.

If the lubricating oil pump housing is damaged, it must be replaced.

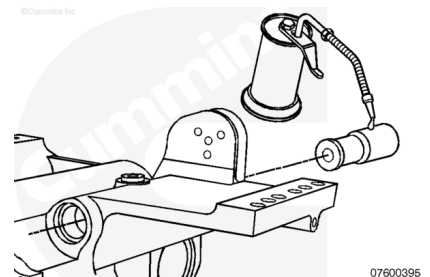


Clean the spool valve and its bore in the oil pump.

Lightly oil the spool valve with clean engine oil.

Install the spool valve. Slide the valve back and forth in the valve bore to make sure no binding or sticking occurs.

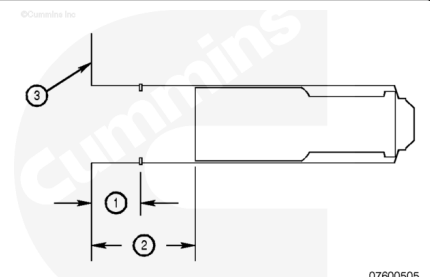
If binding or sticking does occur, determine which part is causing the problem and replace it.



Apply a coat of clean engine oil to the high pressure relief valve spool and install it into the bore. Slide and seat the spool into the bottom of the bore.

Measure and record the distance from the machined front face of the bore to the top of the spool (2).

Subtract the dimension (1), recorded during disassembly, from dimension (2). If the calculated distance is not within the acceptable limit, the lubricating oil pump body must be replaced.



Valve Beat In Specification

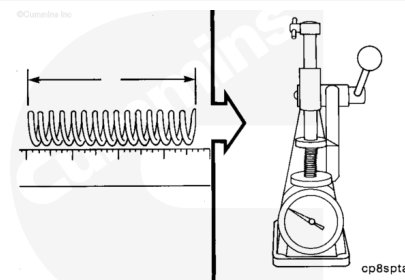
New		Acceptable Limits	
mm	in	mm	in

Valve Beat In Specification

18.237	0.718	Min	18.237	0.718
19.253	0.758	Max	20.65	0.813

Measure the free length of the pressure relief valve spring. The free length is 108 mm [4.25 in].

Use the spring tester, Part Number 3375182, to measure the pressure relief valve spring force at working height, 60 mm [2.33 in].



Lubricating Oil Pump Pressure Relief Valve Spring Force at Working Height

n		lbf
658	MIN	148
725	MAX	163

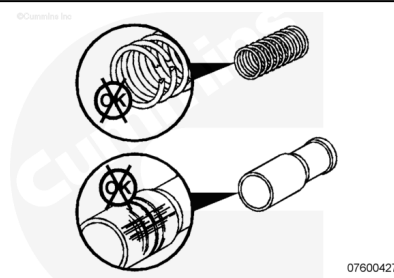
Vertical Position

Inspect the spool valve for scuffing or scoring.

If any damage is found, inspect the high pressure relief valve bore in the oil pump.

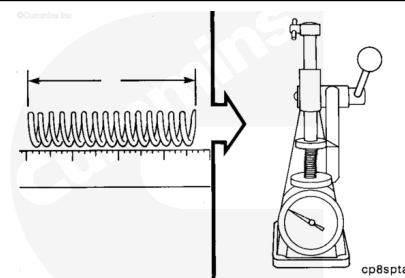
Note : Polished areas on the spool valve indicate normal wear.

Inspect the valve spring for damage.



Measure the free length of the pressure relief valve spring. The free length is 108 mm [4.25 in].

Use the spring tester, Part Number 3375182, to measure the pressure relief valve spring force at working height, 60 mm [2.33 in].



Lubricating Oil Pump Pressure Relief Valve Spring Force at Working Height

n		lbf
658	MIN	148
725	MAX	163

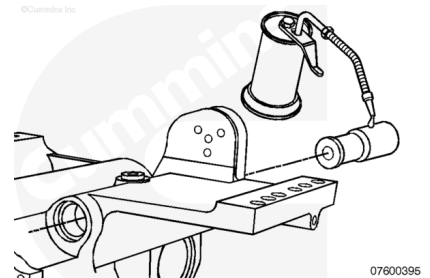
Install

Horizontal Position

Note : Make sure all components of the high pressure relief valve assembly are clean.

Lightly oil the spool valve.

Install the spool valve and spring into the oil pump.



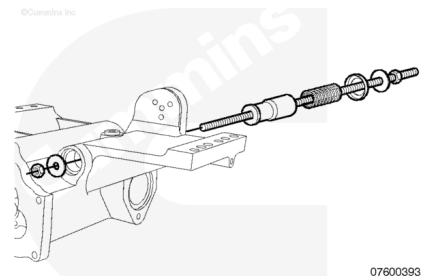
Install the threaded rod through the spool valve and spring.

Install the spring retainer over the threaded rod.

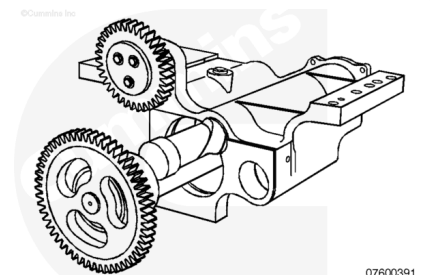
Install the washers and nuts onto the threaded rod.

Tighten the nuts until there is enough clearance to install the snap-ring.

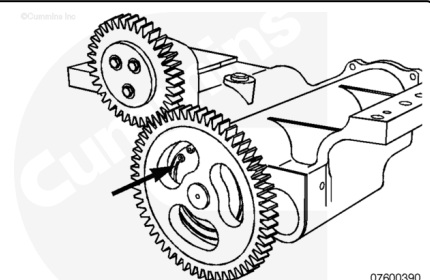
Install the snap-ring and remove the threaded rod.



Slide the oil pump rotor assembly into the oil pump housing.



Install the oil pump main rotor bearing snap-ring.



Install the oil pump gear cover and capscrews.

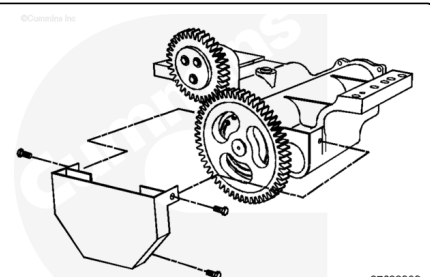
Tighten the capscrews.

Torque Value:

M8 Capscrews 23 n•m [204 in-lb]

Torque Value:

M12 Capscrews 80 n•m [59 ft-lb]



Vertical Position

Note : Make sure all components of the high pressure relief valve assembly are clean.

Lightly oil the spool valve.

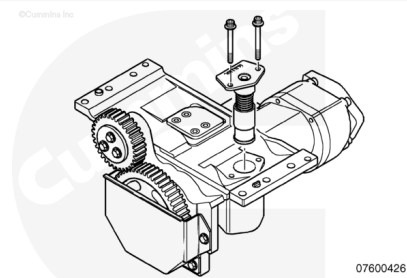
Install the spool valve and spring into the oil pump.

Install the cover and the capscrews.

Tighten the capscrews alternately, i.e. two turns each. This will enable the spring to compress evenly.

Torque Value:

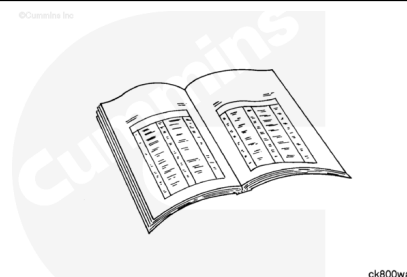
Cover Capscrews 54 n•m [40 ft-lb]



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the lubricating oil pump. Refer to Procedure 007-031 in Section 7.
(/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)
- Operate the engine and check for leaks.